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Electric Power Systems Research 78 (2008) 1561–1567
ELECTRIC POWER SYSTEMS RESEARCH
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A DSP-based active power filter for low voltage distribution systems

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Received 23 July 2007; received in revised form 29 January 2008; accepted 30 January 2008
Available online 20 March 2008

Abstract

The use of nonlinear loads, which inject undesired harmonic currents into low voltage distribution systems, is increasing rapidly. Active power filters are being considered as a potential candidate for solving harmonic problems in order to meet harmonic standards and guidelines. A new digital signal processor (DSP)-based control method for a single-phase active power filter (APF) is presented in this paper. Compared to conventional analog-based methods, the DSP-based solution provides a flexible and cheaper method to control the APF. The proposed scheme employs a carrier-based control that requires less feedback information compared to other reported solutions. Only one current sensor is used to sense the nonlinear load current and two voltage sensors to sense the input supply voltage and the dc bus voltage. The proposed method provides both harmonic elimination and power factor correction. The PSpice simulation and experiments using the DSP-based prototype are made to verify the feasibility of the method.

Keywords: Harmonic distortion; Active filters; Digital signal processors; Power quality; Power filter; Power system harmonics

1. Introduction

Power electronics-based systems, such as static power converters, and motor drives have become more and more popular due to their high efficiency, low cost, flexibility of control and significant energy savings. However, the nonlinear characteristics of these systems are responsible for harmonic pollution of the power distribution system. Harmonics in power distribution systems result in many adverse effects such as: overheating of distribution transformers, low power factor, interference with precision instruments and communication equipment, malfunction of protection equipment, voltage distortion within facilities, overloaded neutral conductors and high levels of neutral to ground voltage. As a result, effective elimination of harmonics from the distribution system has become important both to the utilities and the end users. Passive approaches, although simple, tend to be bulky and expensive at lower harmonic frequencies. Also, a separate LC filter is required for every major harmonic. Moreover the possibility of exciting a resonance condition with the ac system impedance further worsens the situation. Active power filters, which effectively inject compensating currents into the ac lines, have become an attractive alternative to other methods due to its fast response and flexibility. The harmonics of all orders can be compensated by one piece of equipment and better harmonic compensation characteristics can be obtained due to the use of the high controllability and quick response of switch-mode power electronic converters. An integral part of an active compensation device is the detection unit which generates the reference signals. Various methods, e.g. dead beat control, sliding mode control, artificial intelligence, adaptive hysteresis band reactive power control, have been presented in the literature for this purpose [1–5].

In recent years, digital signal processors (DSP) are being used in a variety of applications that require sophisticated control. DSP-based control for active power filters have been proposed in some papers in which general purpose and floating-point DSPs are used [6,7]. A DSP-based single-phase active power filtering solution is proposed in this paper. The proposed technique uses a fixed-point DSP to effectively eliminate system harmonics. The use of fixed point DSP provides a simpler and cheaper solution compared to other existing solutions proposed in literature referenced above. The proposed technique also provides reactive

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doi:10.1016/j.epsr.2008.01.015

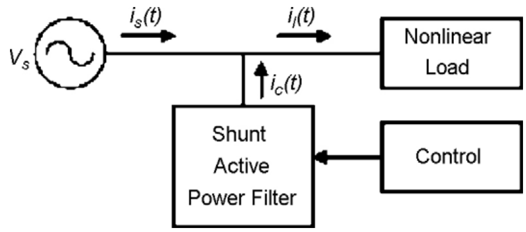

Fig. 1. Active power filter (APF) block diagram.

power compensation or power factor correction. The proposed technique requires only the load current information to estimate the fundamental current information which is then used to calculate the required compensation current using the DSP algorithm. As a result the proposed scheme requires fewer current sensors than other solutions which require both the source current and the load current information [1–5].

2. Basic operation principle of the proposed control method

Fig. 1 shows the compensation principle for a shunt active power filter. The nonlinear load current $i_l(t)$ consists of the fundamental component $i_1(t)$ and harmonic currents $i_h(t)$,

$$i_l(t) = i_1(t) + i_h(t) \tag{1}$$

The APF block supplies compensating currents to cancel the harmonic current components from the source current $i_s(t)$. The compensating current $i_c(t)$ that needs to be supplied by the APF can be obtained by subtracting the fundamental component of the load current from the load current.

$$i_c(t) = i_l(t) - i_1(t) \tag{2}$$

If the active power filter is required to provide reactive power compensation and power factor correction in addition to harmonic compensation, Eq. (2) must be modified. In that case, the APF must supply the reactive component of the fundamental load current in addition to the harmonic currents. Hence, Eq. (2) should be modified such that the compensating current $i_c(t)$ is obtained by subtracting the active component of the load current from the total load current $i_l(t)$.

Fig. 2 shows an active power filter connected in shunt in a single-phase system. The system consists of an ac voltage source, $v_s(t)$, supplying a nonlinear load current $i_l(t)$ to a single-phase diode bridge rectifier circuit. The rectifier circuit represents a typical nonlinear load encountered in modern facilities. The active power filter comprises of a single-phase H-bridge inverter, a capacitor forming a self-regulating dc bus and an inductor connected between the supply and the inverter. The inverter switches are controlled by the DSP controller in order to generate the required compensating current, $i_c(t)$.

The following equations describe the procedure that is used to derive the compensating current, $i_c(t)$.

The source voltage, v_s, in Fig. 2 can be written as,

$$v_s(t) = V_s \sin(\omega t) \tag{3}$$

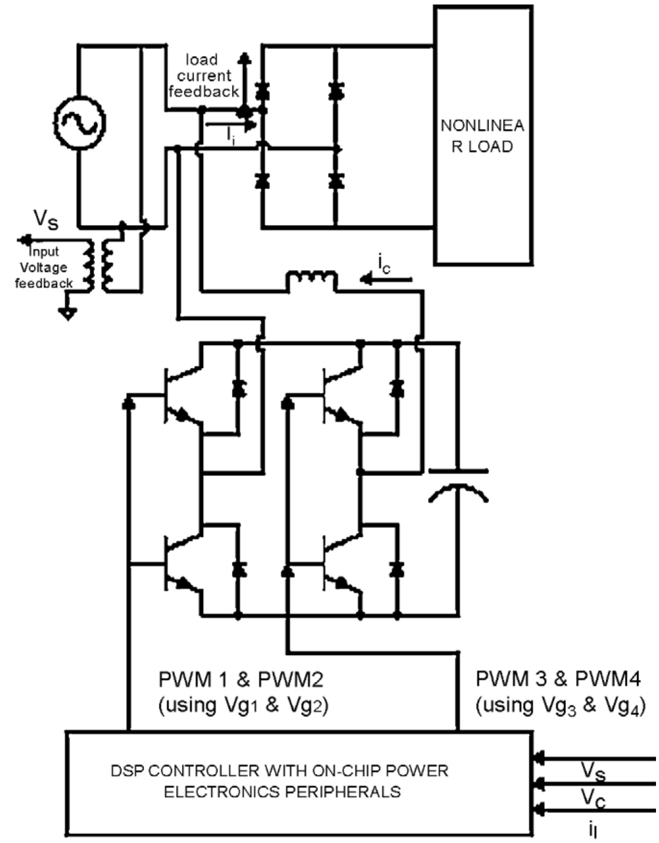

Fig. 2. Block diagram of the proposed active power filter (APF).

The nonlinear load current can be written as:

$$i_l(t) = \sum_{h=1}^{\infty} I_h \sin(h\omega t + \theta_h) \tag{4}$$

where I_h is the peak amplitude of the hth order harmonic in the load current and θ_h is the phase of the hth order harmonic in the load current.Then the load power is

$$\begin{aligned} p_l(t) &= v_s(t) i_l(t) = I_1 V_s \sin(\omega t)\sin(\omega t + \theta_1) \\ &\quad + \sum_{h=2}^{\infty} V_s \sin(\omega t) I_h \sin(h\omega t + \theta_h) = p_s(t) + p_c(t) \end{aligned} \tag{5}$$

In the above equation, $p_s(t)$ includes the power provided by the voltage source and $p_c(t)$ is the harmonic power, which needs to be compensated by the APF. Normally, $p_s(t)$ consists of active power and reactive power. If a unity power factor is desired, the reactive power must be provided by the APF instead of the power source. Therefore, the first term is further decomposed into two terms: the active power and reactive power,

$$\begin{aligned} p_s(t) &= I_1 V_s \sin^2(\omega t)\cos(\theta_1) + I_1 V_s \sin(\omega t)\cos(\omega t)\sin(\theta_1) \\ p_s(t) &= p_a(t) + p_r(t) \end{aligned} \tag{6}$$

where $p_a(t)$ is the active power and $p_r(t)$ is the reactive power. The peak value of the active current is $I_1\cos(\theta_1)$.

The last obstacle is to derive the amplitude. If the load power is integrated over one period, the load power can be expressed

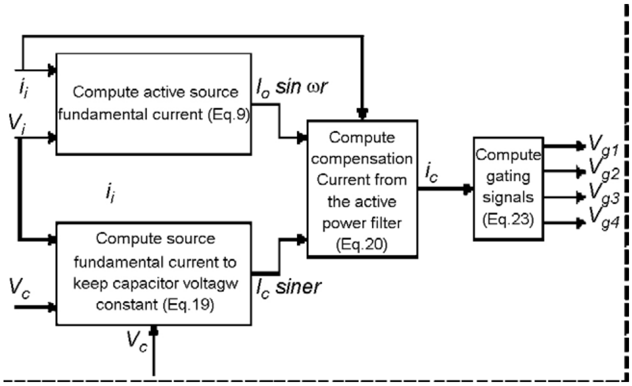

Fig. 3. Block diagram of control scheme.

as:

$$W = \int_0^T V_s \sin(\omega t) \sum_{h=1}^{\infty} I_h \sin(h\omega t + \theta_h)\, dt$$
$$= \frac{V_s T}{2} I_1 \cos(\theta_1) = K_w I_0 \tag{7}$$

where

$$K_w = \frac{V_s T}{2} \tag{8}$$

$$I_0 = I_1 \cos(\theta_1) \tag{9}$$

where T is the time period and K_w is a constant for a fixed source voltage. I_0 is the peak value of the active component of the load current. The compensation current is then given by:

$$i_c(t) = i_l(t) - I_0 \sin(\omega t) \tag{10}$$

The capacitor on the dc side of the APF is an energy storage element. The capacitor voltage will fluctuate depending on the load demand of the system. The following scheme is applied to maintain a constant voltage on the self-regulating dc bus. Assuming that the voltage source provides only the pure sinusoidal current, which is in phase with the voltage, the source current is expressed as

$$i_s(t) = I_s \sin(\omega t) \tag{11}$$

Therefore, the instantaneous source power is,

$$p_s = v_s(t) i_s(t) = \frac{1}{2} V_s I_s - \frac{1}{2} V_s I_s \cos(2\omega t) = P_{sd} + P_{sa} \tag{12}$$

where P_{sd} and P_{sa} are the dc and ac component of the source power, respectively. Similarly, the load power can be decomposed into dc and ac components,

$$p_l(t) = \frac{1}{2} V_s I_1 \cos(\theta_1) + P_{la} = P_{ld} + P_{la} \tag{13}$$

where $p_l(t)$ is the load power and p_{ld} and p_{la} are the dc and ac components of load power, respectively.

So, the compensating power that needs to be supplied by the APF can be derived as follows:

$$p_c(t) = p_s(t) - p_l(t) = (P_{sd} - P_{ld}) + (P_{sa} - P_{la})$$
$$= P_{cd} + P_{ca} \tag{14}$$

where P_{cd} and P_{ca} are the dc and ac component of APF power, respectively. If the reference dc bus voltage is V_c and the capacitor voltage change is ΔV_c, during a period T. The capacitor energy change during a period is

$$\Delta E = \frac{1}{2} C (V_c + \Delta V_c)^2 - \frac{1}{2} C V_c^2 \tag{15}$$

where C is the capacitance value of the dc bus capacitor.

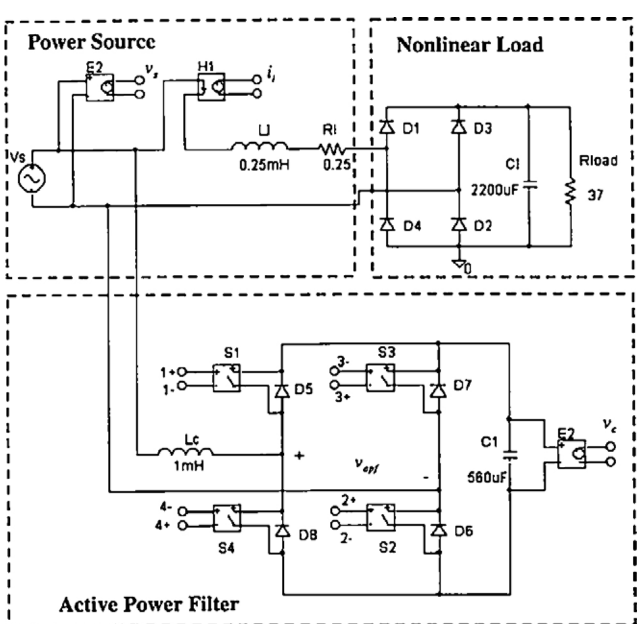

Fig. 4. Simulation circuit schematic in PSpice.

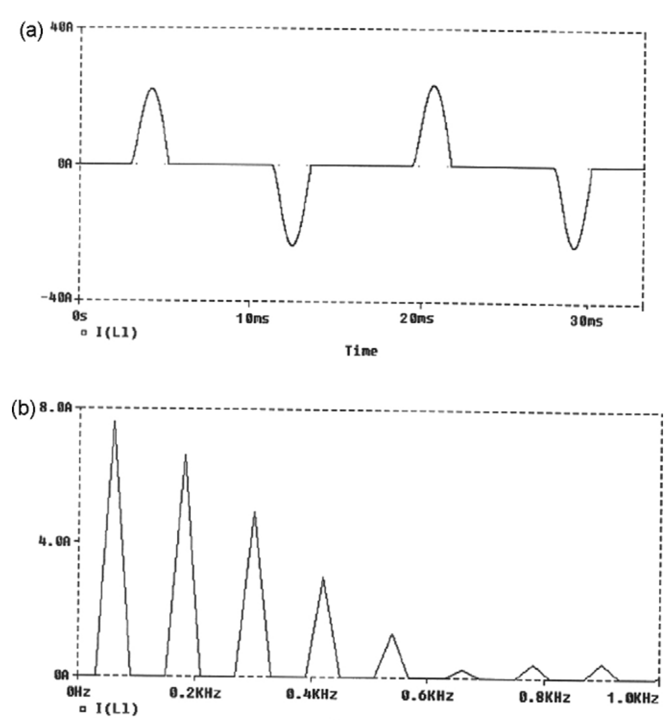

Fig. 5. Load current i_l and its frequency spectrum.

There will be energy exchange between the source and capacitor to keep the capacitor energy storage constant. The dc component P_{cd} of APF power just reflects this kind of energy exchange. So, under the assumption that ΔV_c is small enough to be ignored compared to V_c,

$$\begin{aligned} \Delta E &= C \times V_c \times \Delta V_c \\ \Delta E &= P_{cd} T \\ \Delta V_c &= \frac{V_s T}{2 C V_c}(I_s - I_1 \cos\theta_1) \\ &= K'(I_s - I_0) \end{aligned} \tag{16}$$

where

$$K' = \frac{V_s T}{2 C V_c} \tag{17}$$

From Eq. (16), the following equation is derived as,

$$I_s = I_0 + \frac{1}{K'}\Delta V_c = I_0 + K\Delta V_c = I_0 + I_e \tag{18}$$

$$I_e = \Delta V_c \frac{2 C V_c}{V_s T} \tag{19}$$

where $K = 1/K'$, I_e, is the current keeping the capacitor voltage constant and I_0 is the peak value of the active component in the load current. Then Eq. (10) is modified to,

$$i_c(t) = i_l(t) - I_s \sin(\omega t) \tag{20}$$

Fig. 3 shows the block diagram representation of the DSP-based control scheme. After the required compensation current $i_c(t)$ has been calculated as in Eq. (17), a Pulse Width Modulation (PWM) switching strategy must be adopted to control the inverter switches. With appropriate control the active power filter inverter should generate a current which is equal to the calculated compensating current $i_c(t)$.

The PWM signals for the gate control of the inverter switches are generated by using the calculated compensation current value. A carrier-based unipolar control scheme was adopted. According to the PWM modulation theory, the active power

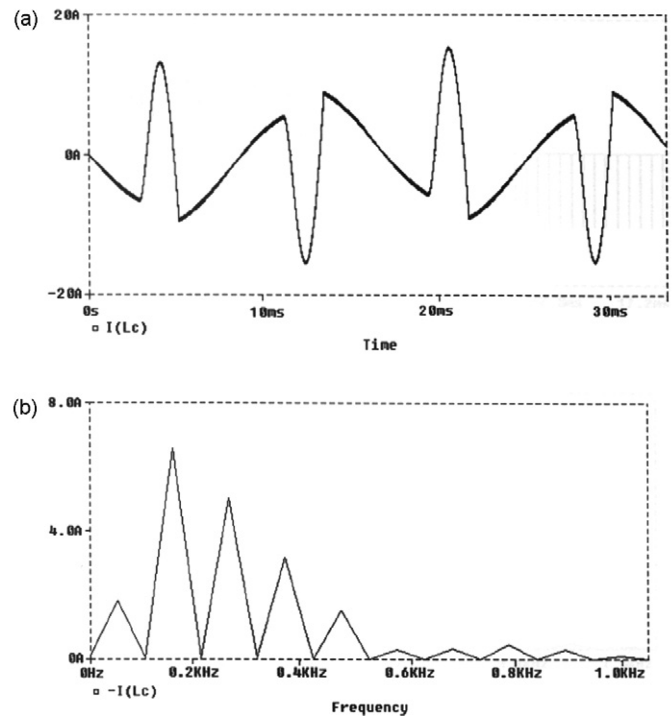

Fig. 7. APF current i_c and its frequency spectrum.

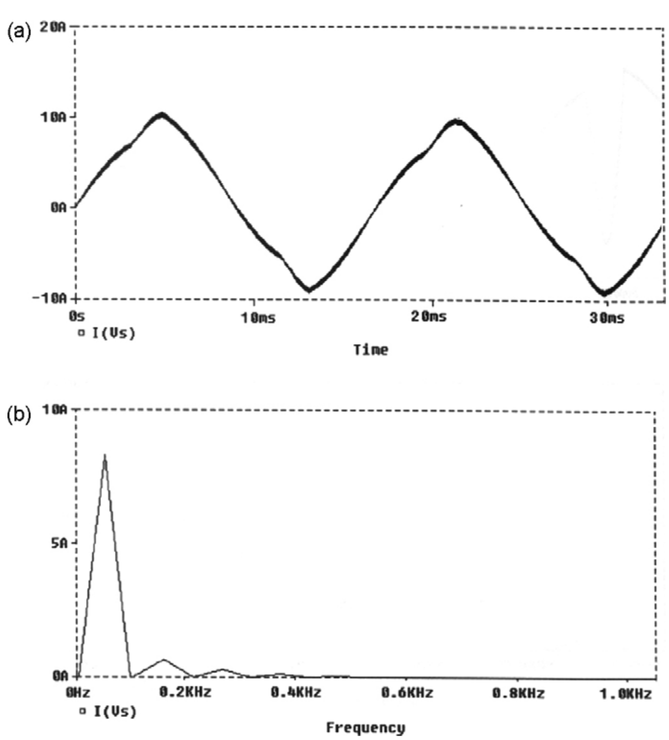

Fig. 6. Source current i_s and its frequency spectrum.

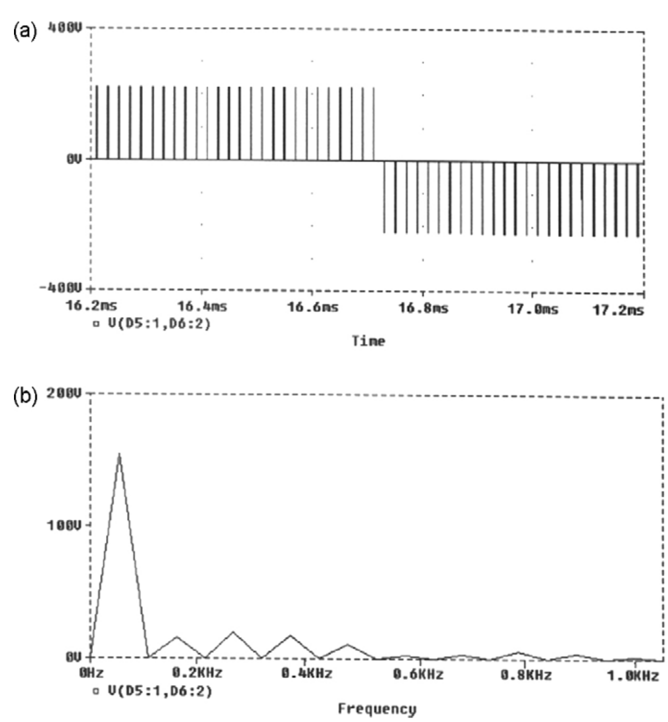

Fig. 8. APF output voltage and its frequency spectrum.

filter output voltage on the ac side is defined by the capacitor voltage v_c, modulation signal v_m and peak carrier voltage V_{tri} as,

$$v_{apf} = \frac{v_m}{V_{tri}} V_c \tag{21}$$

The compensation current i_c and the active power filter output voltage v_{apf} could be related as:

$$i_c = \frac{1}{L_c} \int (V_{apf} - v_s)\,dt \tag{22}$$

where L_c represents the inductance between the source and the inverter and v_s is the source voltage. From Eqs. (21) and (22), the modulation signal can be expressed in terms of the calculated compensation current as,

$$v_m = \frac{L_c d/dt(i_c) + v_s}{V_c/V_{tri}} \tag{23}$$

3. PSpice simulation

The proposed control method was simulated using PSpice. The simulation circuit schematic is shown in Fig. 4. Three circuit variables i_l, v_s and v_c, are sensed to calculate the compensation current i_c and hence the control signals for the switches. Before the active power filter is connected to the circuit the source current and the load current are the same. The load current i_l and its frequency spectrum are shown in Fig. 5. From the frequency spectrum shown in Fig. 5(b), it is obvious that the load is nonlinear since harmonic components (180 Hz, 300 Hz, etc.) are present in addition to the fundamental 60 Hz component. From Fig. 6 it can be seen that the source current now looks more sinusoidal and that the harmonic components present in the load current have been removed from the source current. The load current still remains the same as shown in Fig. 5. However, the APF provides all the harmonic current requirement of the load as can be seen in Fig. 7 which shows the compensation current

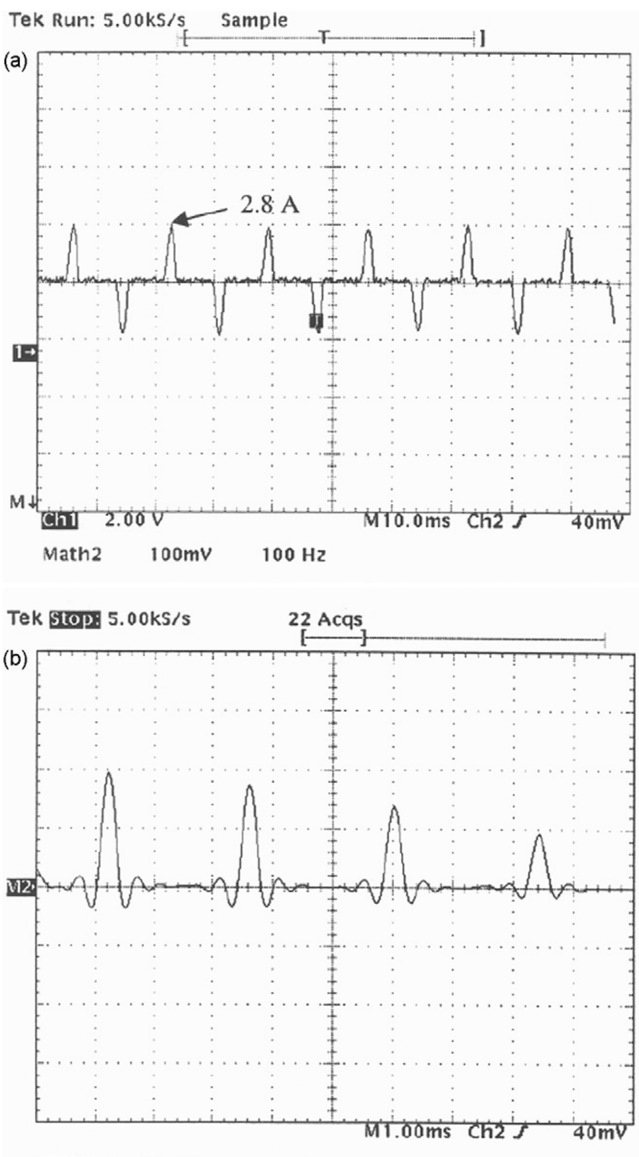

Fig. 9. Load current i_l and its frequency spectrum.

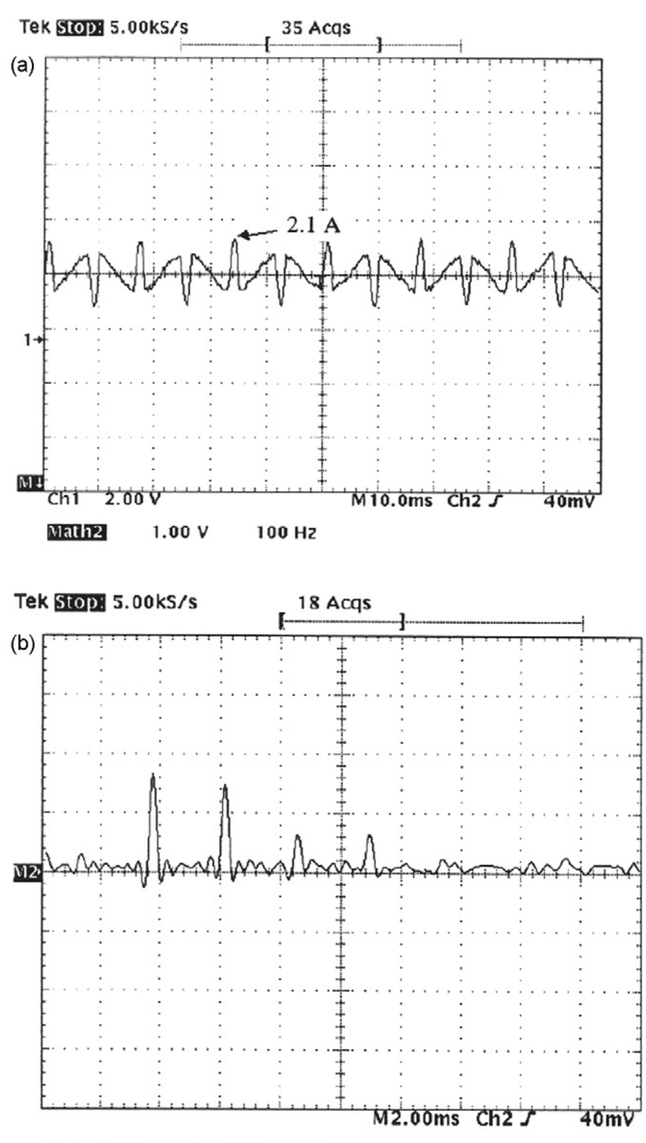

Fig. 10. APF current i_c and its frequency spectrum.

i_c and its frequency spectrum. A small amount of fundamental current is also found in the frequency spectrum of i_c shown in Fig. 7(b), which represents the power loss in the APF due to frequent switching of the switches. Fig. 8 shows the APF output voltage, v_{apf}.

4. DSP implementation and experimental results

The hardware setup was implemented as shown in the circuit diagram of Fig. 2. The DSP controller TMS320F240 from Texas Instruments was utilized to implement the proposed active power filtering method. It is a fixed point digital signal processor, which is different from general purpose floating-point DSPs that are used in most applications. Fixed point processors are less expensive than the floating point ones, however, more efforts are required to develop appropriate scaling factors to eliminate the effects of truncation or overflow. A diode bridge rectifier with resistive load acted as the nonlinear load for the system. A hall-effect current sensor was used to measure the load current, i_l. The source voltage v_s and capacitor voltage v_c were sensed using voltage divide circuits before connecting to the analog-to-digital channels of DSP. The calculation of reference compensating current i_c and generation of PWM control signals were completed on the DSP board. Four PWM channels (PWM1 through PWM4) control the voltage source inverter. These four PWM channels are grouped in two pairs (PWM 1&2, and PWM 3&4). Two compare registers are associated with each PWM channel pair. The compare register values are updated to obtain the proper PWM output. The on-chip software programmable, dead band module provides sufficient dead time to avoid shoot through faults. The on chip analog to digital conversion module was utilized to measure the required current and voltages.

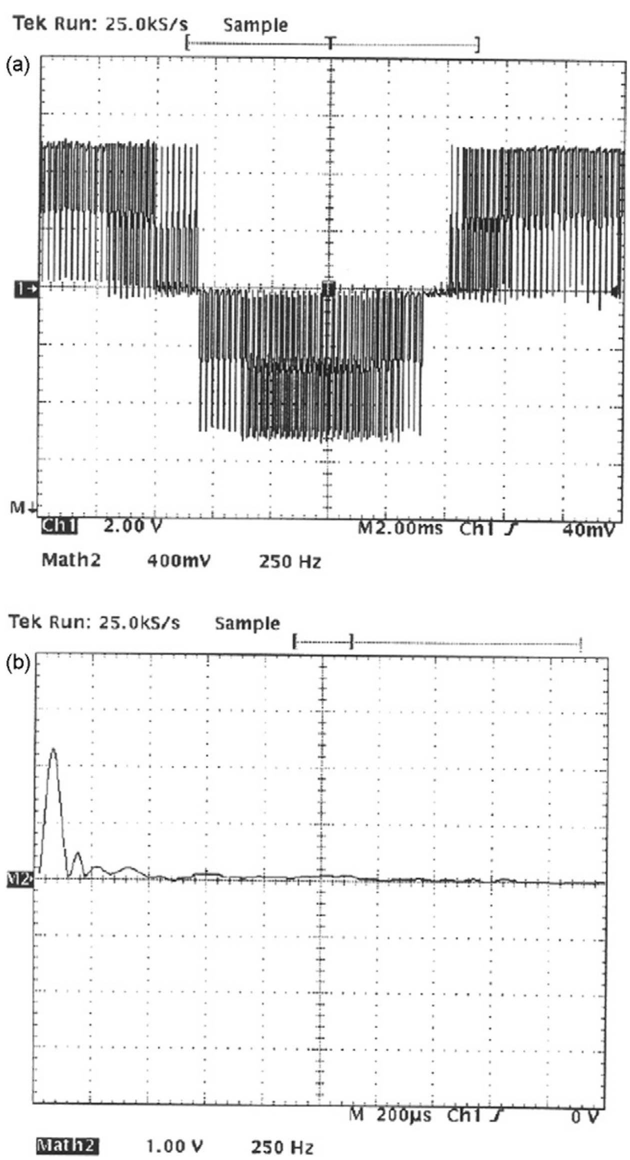

Fig. 11. APF output voltage and its frequency spectrum.

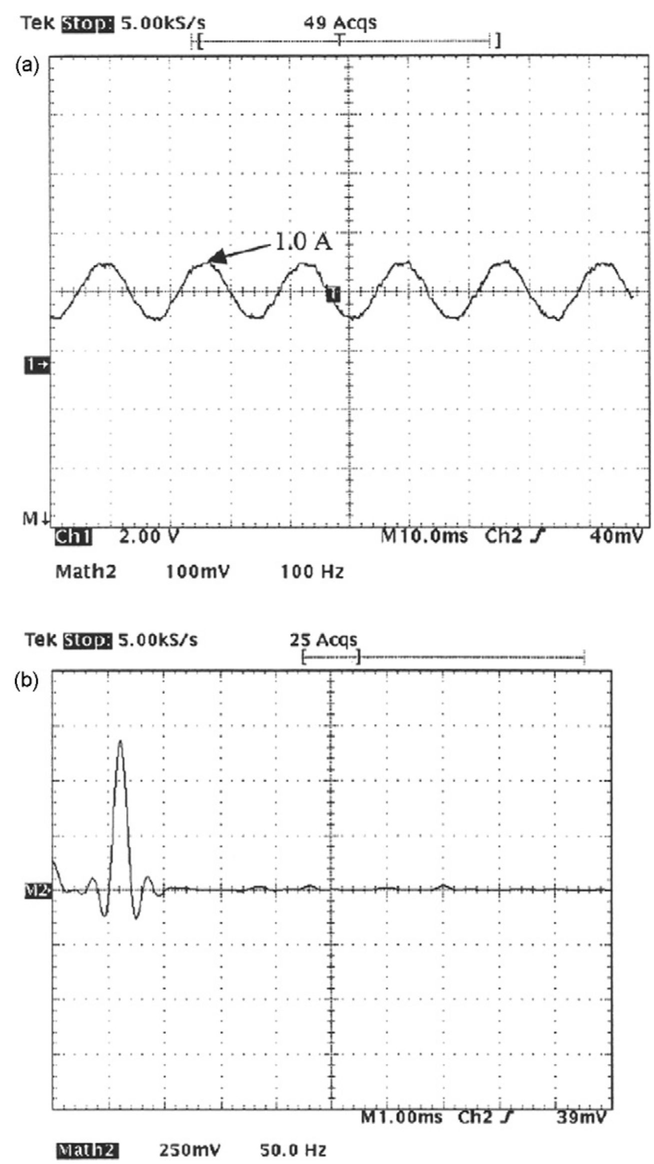

Fig. 12. Source current and its frequency spectrum after compensation.

The maximum measurement range of the load current sensor was 3 A. A 32 Ω resistor was chosen as the rectifier load to make sure the load current fell into the valid measurement range. The sensed source voltage transformer ratio was 63.5, the load current transformer ratio was 1.5 and the capacitor voltage transformer ratio was 13.

The experimental setup of the system yielded the following experimental results. Fig. 9 shows the load current i_l and its frequency spectrum. The on board digital-to-analog converter (DAC) module was used to output the sensed load current to the oscilloscope for display. It is obvious that the load current is non-sinusoidal and consists of the 60 Hz fundamental component along with lower order harmonics like the third harmonic (180 Hz), fifth harmonic (300 Hz), seventh harmonic (420 Hz), etc. The fundamental component in the load current was estimated using the proposed method. After calculation it was output through the DAC channel of the DSP board. The compensating current i_c was obtained after subtracting the estimated fundamental current from the load current. The compensating current $i_c(t)$ is shown in Fig. 10. The compensating current was converted to the gate control signals for the switches using the proposed method (Eqs. (17)–(20). Fig. 11 shows the active power filter output voltage v_{apf} and its corresponding frequency spectrum. Fig. 12 shows the source current and its frequency spectrum after compensation. It is obvious from Fig. 12 that the proposed scheme successfully compensates and achieves a near sinusoidal input line current. The experimental results were found to be comparable with the simulation results.

5. Conclusions

A DSP-based active power filter for a single-phase system is presented. The scheme utilizes the on-chip power electronics peripherals of a DSP controller. This minimizes the overall chip count of the system. The proposed scheme requires less feedback information compared to other solutions. The technique relies on only one current sensor used to sense the nonlinear input current and two voltage sensors.

References

[1] J. Matas, L. Garcia de Vicuna, J. Miret, J.M. Guerrero, M. Castilla, Feedback linearization of a single-phase active power filter via sliding mode control, Power Electron. IEEE Trans. 23 (1) (2008) 116–125.

[2] R.E. Shatshat, M.M.A. Salama, M. Kazerani, Artificial intelligent controller for current source converter-based modular active power filters, Power deliv. IEEE Trans. 19 (3) (2004) 1314–1320.

[3] K. Nishida, M. Rukonuzzman, M. Nakaoka, Advanced current control implementation with robust deadbeat algorithm for shunt single-phase voltage-source type active power filters, Electric Power Applic. IEE Proceedings 151 (3) (8 May. 2004) 283–288.

[4] T. Toshihiko, H. Eiji, I. Norio, O. Yasushi, Y. Masayoshi, A novel real-time detection method of active and reactive currents for single-phase active power filters, Power Electronics Specialists Conference, 2007 (PESC 2007), IEEE, 17–21 June, 2007, pp. 2933–2938.

[5] A.N. Jog, N.G. Apte, An Adaptive Hysteresis Band Current Controlled Shunt Active Power Filter, Compatibility in Power Electronics, 2007 (CPE '07), May 29 2007–June 1 2007, pp.1–5.

[6] D. Lindeke, S.A. Mussa, F.P. de Souza, I. Barbi, Single phase active power filter controlled with a digital signal processor – DSP, Power Electronics Specialists Conference, 2004 (PESC 04), 2004 IEEE 35th Annual vol. 2, 20–25 June 2004, pp.1024–1027.

[7] G. Sincy, V. Agarwal, A DSP-based control algorithm for series active filter for optimized compensation under nonsinusoidal and unbalanced voltage conditions, IEEE Trans. Power Deliv. 22 (n 1) (2007) 302–310.

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